

Reasoning Reference: Logical Puzzles

Educational Series

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1 Introduction

Logical puzzles require the organization of scattered information into a structured format (usually a table or a diagram). The goal is to match multiple variables (e.g., Names, Professions, Cities, Colors) based on a set of positive and negative constraints.

2 Types of Puzzles

2.1 Classification Puzzles (Grid Method)

These involve matching items from different categories.

- **Method:** Create a table with variables as headers. Use tick marks (✓) for confirmed matches and crosses (×) for ruled-out matches.
- **Rule of One:** In most standard puzzles, if one cell in a row/column is a ✓, all other cells in that row/column must be ×.

2.2 Seating Arrangements

- **Linear Arrangement:** Objects are placed in a row. Pay attention to "Immediate Left/Right" vs. just "Left/Right."
- **Circular Arrangement:** Objects are placed around a circle.
 - *Facing Inward:* Clockwise is Left; Anti-clockwise is Right.
 - *Facing Outward:* Clockwise is Right; Anti-clockwise is Left.

2.3 Comparison Puzzles

Involve ranking based on height, weight, marks, etc.

- **Method:** Use inequality symbols ($>$, $<$).
- **Example:** $A > B = C > D$.

3 The Solving Algorithm

1. **Identify Variables:** List the sets (e.g., 5 people, 5 cars, 5 colors).
2. **Fix a Base:** Choose the most static variable (usually Names or Days of the week) as the primary column.
3. **Direct Information:** Fill in the certain/positive facts first.
4. **Negative Constraints:** Note down what *cannot* happen (e.g., "A is not the Doctor") next to the table.
5. **Linked Information:** Look for clues that connect two or more variables.
6. **Final Fill:** Use the process of elimination for remaining gaps.

4 Solved Example: Seating

Problem: Five friends (P, Q, R, S, T) are sitting in a row facing North. S is between T and Q. Q is to the immediate left of R. P is at the extreme left. Who is in the middle?

Solution:

1. P is at extreme left: P _ _ _ _
2. Q is to the immediate left of R: This means they are a pair QR.
3. S is between T and Q: This means T S Q.
4. Combine: Since P is far left, and we have the sequence T S Q R:
5. Full Row: P T S Q R
6. **Answer:** S is in the middle.

5 Solved Example: Grid

Problem: A, B, and C play different games (Cricket, Football, Tennis). A does not play Cricket. B plays Tennis. Which game does C play?

Solution:

	Cricket	Football	Tennis
A	×	✓	×
B	×	×	✓
C	✓	×	×

Since B plays Tennis, A and C cannot. Since A doesn't play Cricket or Tennis, A must play Football. Thus, C must play Cricket.